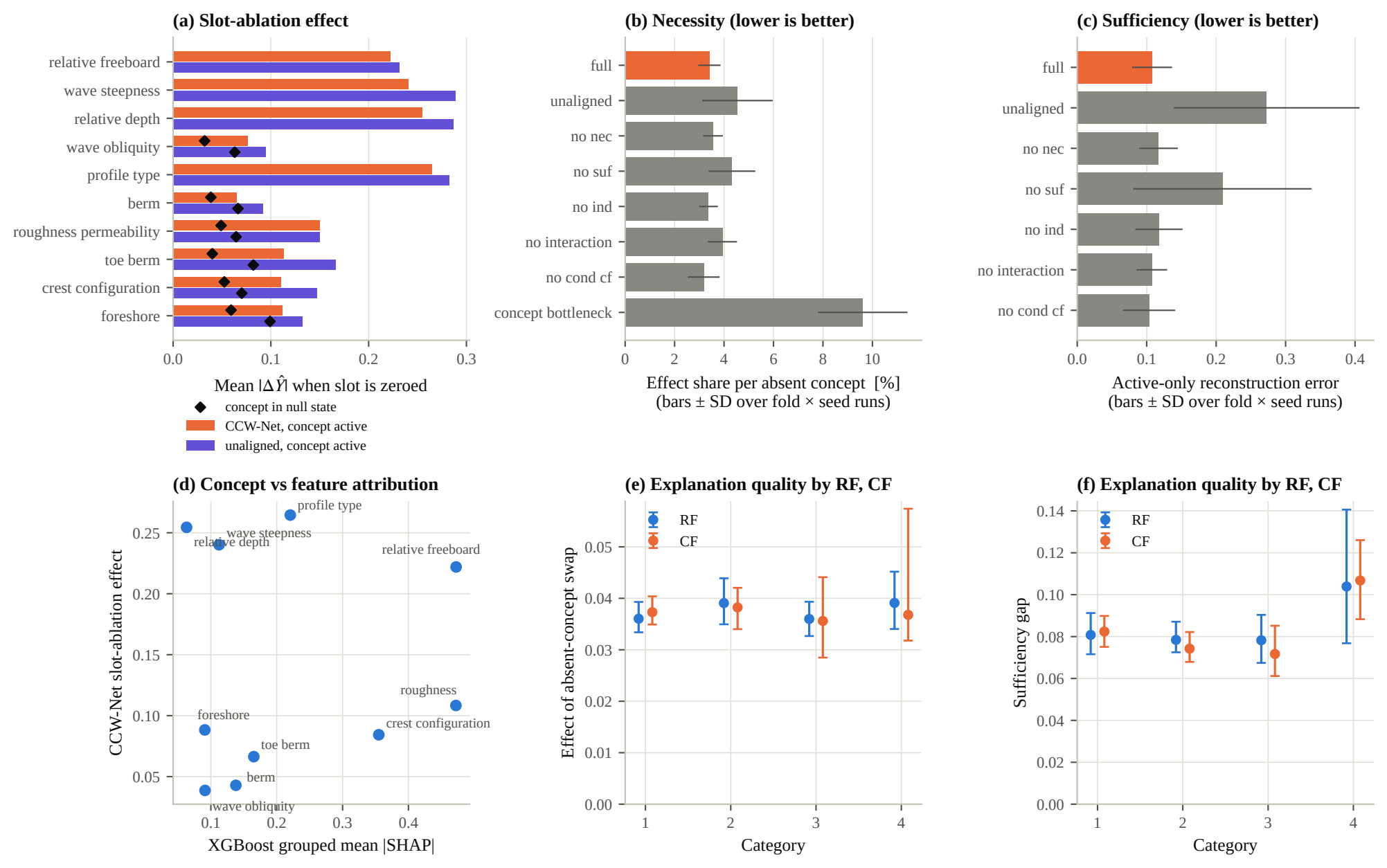
(a) Slot-ablation effect
relative freeboard
wave steepness
relative depth
wave obliquity
profile type
berm
roughness permeability
toe berm
crest configuration
foreshore
0.0
0.1
0.2
0.3
Mean |ΔŶ| when slot is zeroed
concept in null state
CCW-Net, concept active
unaligned, concept active
(b) Necessity (lower is better)
full
unaligned
no nec
no suf
no ind
no interaction
no cond cf
concept bottleneck
0
2
4
6
8
10
Effect share per absent concept [%]
(bars ± SD over fold × seed runs)
(c) Sufficiency (lower is better)
full
unaligned
no nec
no suf
no ind
no interaction
no cond cf
0.0
0.1
0.2
0.3
0.4
Active-only reconstruction error
(bars ± SD over fold × seed runs)
(d) Concept vs feature attribution
profile type
wave steepness
relative depth
relative freeboard
roughness
foreshore
crest configuration
toe berm
berm
wave obliquity
0.25
0.20
0.15
0.10
0.05
CCW-Net slot-ablation effect
0.1
0.2
0.3
0.4
XGBoost grouped mean |SHAP|
(e) Explanation quality by RF, CF
RF
CF
0.05
0.04
0.03
0.02
0.01
0.00
Effect of absent-concept swap
1
2
3
4
Category
(f) Explanation quality by RF, CF
RF
CF
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
Sufficiency gap
1
2
3
4
Category